Tanqiu Jiang

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Education

Stony Brook University — Stony Brook, NY
Ph.D. in Computer Science — *Aug 2023 – Present*

Pennsylvania State University — State College, PA
Ph.D. in Informatics (transferred to Stony Brook) — *Aug 2022 – May 2023*

University of Rochester — Rochester, NY
Master of Science in Data Science — *Aug 2020 – Dec 2021*

Lehigh University — Bethlehem, PA
Bachelor of Science in Computer Engineering (Minor: Data Science) — *Aug 2016 – May 2020*

Publications

Under Review

[1] **Jiang, T.**, Bai, M., Pappas, N., Qi, Y., & Swamy, S. Cross-Modal Content Optimization for Steering Web Agent Preferences. (**Submitted to ICML 2026**). [pdf]

[2] **Jiang, T.**, Wang, Y., Liang, J., & Wang, T. AgentLAB: Benchmarking LLM Agents against Long-Horizon Attacks. (**Submitted to ICML 2026**). [pdf]

[3] Liang, J., Wang, Y., **Jiang, T.**, & Wang, T. RASA: Routing-Aware Safety Alignment for Mixture-of-Experts Models. (**Submitted to ICML 2026**). [pdf]

[4] Liang, J., **Jiang, T.**, Wang, Y., Zhu, R., Ma, F., & Wang, T. AutoRAN: Weak-to-Strong Jailbreaking of Large Reasoning Models. (**Submitted to ACL 2026**). [pdf]

[5] Zhu, R., Wang, Y., **Jiang, T.**, Liang, J., & Wang, T. Self-Improving Model Steering. (**Submitted to ICML 2026**). [pdf]

Conference Papers

[6] **Jiang, T.**, Liang, J., Zhu, R., Zhou, J., Ma, F., & Wang, T. (2026). Dynamic Token Reweighting for Robust Vision-Language Models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR 2026**). [pdf]

[7] Liang, J., Wang, Y., Li, C., Zhu, R., **Jiang, T.**, Gong, N., & Wang, T. (2026). GraphRAG Under Fire. In *IEEE Symposium on Security and Privacy* (**S&P 2026**). [pdf]

[8] **Jiang, T.**, Wang, Z., Liang, J., Li, C., Wang, Y., & Wang, T. (2025). RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction. In *International Conference on Learning Representations* (**ICLR 2025**). [pdf]

[9] **Jiang, T.**, Li, C., Ma, F., & Wang, T. (2025). RAPID: Retrieval Augmented Training of Differentially Private Diffusion Models. In *International Conference on Learning Representations* (**ICLR 2025**). [pdf]

[10] **Jiang, T.**, Li, Y., Lin, H., Ruan, Y., & Woodruff, D. P. (2020). Learning-Augmented Data Stream Algorithms. In *8th International Conference on Learning Representations* (**ICLR 2020**), Addis Ababa, Ethiopia. [pdf]

[11] **Jiang, T.**, Bendre, S. K., Lyu, H., & Luo, J. (2021). From Static to Dynamic Prediction: Wildfire Risk Assessment Based on Multiple Environmental Factors. In *2021 IEEE International Conference on Big Data* (**IEEE Big Data**). [pdf]

[12] **Jiang, T.** & Xiong, Z. (2021). Rule-Based Approach to the Automatic Detection of Individual Tree Crowns in RGB Satellite Images. In *2021 IEEE International Conference on Computer Science, Artificial Intelligence and Electronic Engineering (IEEE-CSAIEE)*, pp. 132–135. [pdf]

[13] Wang, X., **Jiang, T.**, & Cai, H. II. (2021). Human Epithelial-2 Cell Image Classification Using Deep Unsupervised Learning and Gradient Boosting Trees. In *Proc. SPIE 11601, Medical Imaging 2021*. [pdf]

Journal Papers

[14] Tian, X., Liu, Y., Zhang, K. L., Litt, S., Shin, K., & **Jiang, T.** (2026). Do Artificial Intelligence Clients Speak Like Human Clients? Exploring GPT-4's Content-Level Performance in Counseling Role-Play. *Journal of Counseling & Development*, 104(1), 14–26.

[15] Tian, X., Liu, Y., Litt, S., Shin, K., & **Jiang, T.** (2025). Artificial Clients, Genuine Interactions? Assessing ChatGPT-4's Process-Level Performance in Clinical Simulations Using Qualitative Content Analysis. *Counselling and Psychotherapy Research*, 25(4), e70063.

Preprints

[14] Liang, J., **Jiang, T.**, Wang, Y., Zhu, R., Ma, F., & Wang, T. (2025). AutoRAN: Weak-to-Strong Jailbreaking of Large Reasoning Models. *arXiv e-prints, arXiv:2505.10846*. [pdf]

[15] Zhu, R., Wang, Y., **Jiang, T.**, Liang, J., & Wang, T. (2025). Self-Improving Model Steering. *arXiv preprint arXiv:2507.08967*. [pdf]

Experience

Amazon Seattle, WA

Applied Scientist Intern, Agentic AI Lab *May 2025 – Sep 2025*

- Developed cross-modal adversarial attack framework to identify vulnerabilities in VLM-based web agents, achieving up to 71% preference manipulation rate across GPT-4.1, Qwen-2.5VL, and Pixtral-Large models.
- Implemented PGD-based visual perturbation attacks exploiting CLIP encoder vulnerabilities, achieving $>$50% black-box transfer success while maintaining imperceptibility (ϵ=8/255).
- Designed iterative textual refinement algorithms leveraging RLHF-induced biases to steer agent preferences through semantically plausible content modifications.
- Conducted comprehensive evaluation on ControlNet-100k and VisualWebArena benchmarks, demonstrating 3.5$\times$ improvement over baseline methods while evading detection.

Stony Brook University Stony Brook, NY

Research Assistant *Aug 2023 – Present*

- Conduct advanced research in computer science under the supervision of Dr. Ting Wang, focusing on enhancing the robustness and privacy of machine learning models.
- Lead and manage two ongoing research projects:
  - **RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction**
  - **RAPID: Retrieval Augmented Training of Differentially Private Diffusion Models**
- Coordinate with a multidisciplinary team to design experiments, analyze data, and draft research manuscripts for publication.
- Present research findings at internal seminars and collaborate with external researchers to refine methodologies and approaches.

Pennsylvania State University State College, PA

Research Assistant *Aug 2022 – May 2023*

- Led a research project developing "Unlearnable Examples" aimed at minimizing the performance of contrastive learning models by introducing adversarial noise into training images.
- Attended and presented in weekly individual and group meetings with Dr. Ting Wang, Dr. Dongwon Lee, and lab mates.

University of Rochester, Goergen Institute for Data Science Rochester, NY

Teaching Assistant *Jan 2021 – Dec 2021*

- Held office hours averaging over 2 hours weekly to assist students with Python, Linux, SQL, PySpark, and R.
- Collaborated with Professors Lloyd Palum and Brendan Mort to test and refine course materials on Databricks.
- Conducted review sessions to reinforce course material and prepare students for assessments.
- Graded assignments and projects, providing constructive feedback to students.

Vista Lab, University of Rochester Rochester, NY

Student Researcher *Dec 2020 – Dec 2021*

- Led a research project on analyzing and predicting California Wildfire Incidents under the mentorship of Prof. Jiebo Luo and Hanjia Lyu.
- Integrated multiple environmental factors to assess wildfire risk using Logistic Regression, SVM, Neural Networks, etc.
- Utilized LSTM/RNN for time-series analysis to perform dynamic risk predictions.
- Co-authored a research paper published in the IEEE International Conference on Big Data.

IEEE Big Data 2021 Conference Remote

Student Volunteer / Cohost *Dec 2021*

- Cohosted the special session "S29: Contrastive Learning" alongside the session chair.
- Ensured speakers' presentations adhered to the schedule and were displayed correctly.

Competitions

Kaggle: Google Landmark Retrieval 2020, Awarded Silver Medal (Top 3%, 16th/541).

Grants and Scholarships

Graham Endowed Fellowship ($4,000), Pennsylvania State University, Aug 2022

NSF Student Travel Grant ($500), IEEE-Big Data Conference, Dec 2021